Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/04/2005
PI Name:	West, John B M.D., Ph.D.		
Project Title:	The Effects of EVA and Long-Term Exposure to Microgravity on Pulmonary Function		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92093-0623	**Congressional District:**	49
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	03/02/1999	**End Date:**	12/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS Increments 3,4,5,6		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Prisk, Gordon Kim (University of California - San Diego) Paiva, Manuel (Universite Libre de Bruxelles) Darquenne, Chantal (University of California - San Diego)		
Grant/Contract No.:	NCC9-168		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to use the earliest Utilization Phase of the International Space Station to study the effects of Extra-Vehicular Activity (EVA) and long-term exposure to microgravity on the pulmonary system. From our previous studies of the lung in shorter periods of microgravity (up to 14 days) on Spacelabs SLS-1, SLS-2, and D-2, we know that the lung is profoundly affected by the removal of gravity. We will extend this research in three specific areas: (1) Changes in ventilation- perfusion inequality in the lung and closing volume caused by the effects of venous gas microemboli filtered by the pulmonary vasculature. These microbubbles are likely to occur in the hypobaric environment of EVA, and the effects will be measured using changes in the intra-breath respiratory exchange ratio test (intra-breath R), and in the distribution of pulmonary perfusion measured by the hyperventilation breathhold technique; (2) Changes in small airways function measured by forced spirometry and by the intra-breath R test and changes in the subdivisions of lung volumes measured by slow spirometry. Such changes might be expected due to possible structural changes in lung tissue promoted by long-term changes in pulmonary vascular filling; (3) Changes in respiratory muscle strength measured by forced spirometry, and by maximum inspiratory and expiratory pressures. Such changes might be expected in long-term ìG exposure as the weight of the rib cage and shoulder girdle are removed. These tests, performed at numerous points throughout the flights and before and after EVA, have been chosen because they are simple to perform and require only cabin air for their performance, and are therefore compatible with the operational constraints of the early phase of the ISS. Nevertheless, they provide a comprehensive picture of the effects of EVA and long-duration exposure to ìG on the pulmonary system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Careful monitoring of the lung function of astronauts exposed to microgravity for an extended period of time may ultimately prove useful in providing a better understanding of the way in which gravity affects the lung on earth. The proposed studies will have relevance in providing future space travelers with better information on how microgravity will affect their lungs. For those astronauts who must perform EVA, this study will provide valuable information on the detrimental consequences of the EVA suits and protocols currently in use.
Task Progress:	In the past year, the PuFF experiment has completed the contract. The experiment has been operated successfully on all occasions without major malfunctions or equipment failures. Data quality is excellent. Scheduling of the experiment, for the most part met experiment requirements, with some difficulties during I3. The numbers of EVAs performed has been less than expected, limiting the data collection associated with the objective pertaining to the effects of EVA on lung function, however some post-EVA data were collected on 8 crewmembers. In addition, operational constraints have precluded post-EVA data collection on the day of EVA as originally desired. Data analysis is in progress and the final reporting of results (in the form of peer-reviewed publications) is in progress. To-date, results have only been reported in abstract form.
Bibliography Type:	Description: (Last Updated: 01/19/2006)
Abstracts for Journals and Proceedings	PRISK,G.K.; FINE, J.M.;COOPER, T.K.;WEST,J.B. "VENTILATION-PERFUSION RATIO INEQUALITY IN THE LUNG IS SLIGHTLY INCREASED ONE DAY FOLLOWING EXTRAVEHICULAR ACTIVITY FROM THE ISS" Aerospace Medical Association Aviation Space and Environmental Medicine , May-2004
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